

Gaussian and Non-Gaussian Photonic States

Wigner Positivity, Moment Diagnostics, and Numerical Convergence

Boris Kiefer
New Mexico State
University
bkiefer@nmsu.edu
[GitHub](#)
[LinkedIn](#)



Purpose: Use an interactive, validated simulator to connect Fock-state structure, quadrature statistics, Wigner functions, Gaussianity, and numerical convergence for vacuum, coherent, squeezed, Fock, and mixed photonic states.

Teaching note • 9 September 2026 • Student-facing guide for a validated Jupyter/QuTiP photonic-state simulator.

Abstract

Gaussian states are central to continuous-variable photonics, but a positive Wigner function is not, by itself, a universal test of Gaussianity. This teaching simulator makes that distinction concrete by placing the Wigner function, Fock-state probabilities, and a rotated-quadrature moment hierarchy side by side. Students can compare vacuum, coherent, squeezed-vacuum, Fock, and vacuum–single-photon mixed states; vary coherent amplitude and phase, squeezing in dB, photon number, and mixture weight; and inspect how Gaussianity and Wigner negativity respond. The control layer exposes numerical choices that are usually hidden, including Hilbert-space dimension, phase-space extent, grid resolution, quadrature sampling, moment-axis range, and the numerical cutoff used to call a Wigner value negative. The result is a compact laboratory for distinguishing physical non-Gaussianity from numerical artifacts and for learning why pure-state Wigner positivity, mixed-state Wigner positivity, and Gaussianity are related but not identical concepts.

Impact

Impact hypothesis. A side-by-side physics-and-control interface can help photonics students move beyond visual recognition of “Gaussian-looking” phase-space distributions. The simulator ties qualitative plots to quantitative moment tests, purity, and convergence checks, making state classification, squeezing conventions, and numerical reliability directly testable rather than implicit.

Customer discovery

Photonics students	Quantum-optics instructors	Laboratory trainees / research	Programs and workforce training
Need a direct connection between Fock expansions, quadratures, Wigner functions, squeezing, and Gaussianity rather than treating them as separate topics.	Need a compact activity that supports prediction, parameter variation, and interpretation without requiring students to build the numerical machinery first.	Need practice distinguishing physical signatures from finite-basis and finite-grid artifacts and documenting convergence choices.	Can use the notebook as a portable module for quantum photonics, continuous-variable quantum information, and computational quantum-optics training.

1. Learning goals and recommended use

This notebook is designed for upper-level undergraduate, beginning graduate, or research-onboarding use in quantum optics and photonics. It assumes that students know basic linear algebra and the harmonic oscillator, but it does not assume prior familiarity with phase-space quasiprobabilities.

After completing the activities, students should be able to:

1. distinguish Gaussianity from Wigner positivity and from purity;
2. relate coherent and squeezed states to their Fock-state expansions;
3. use rotated quadrature moments to test Gaussian consistency;
4. identify Wigner negativity in Fock states and understand its numerical resolution;
5. explain why a mixed state can be non-Gaussian while retaining an everywhere nonnegative Wigner function;
6. convert squeezing in dB to the squeezing parameter r ;
7. perform a basis-convergence study by varying the Hilbert-space dimension N ;
8. separate a physical conclusion from a numerical threshold or truncation choice.

A productive classroom pattern is *predict* $\rightarrow$ *simulate* $\rightarrow$ *compare* $\rightarrow$ *explain*. Before moving a control, students should state what they expect to happen to the Wigner function, Fock distribution, and moment residuals.

2. Simulator overview

The simulator generates three synchronized views for each selected state:

- (a) the Wigner function $W(x, p)$;
- (b) Fock probabilities $P_n = \langle n | \rho | n \rangle$;
- (c) normalized deviations of central quadrature moments from Gaussian expectations.

The interface can display one state at a time or a direct multi-row comparison. The standard comparison set contains coherent, squeezed-vacuum, Fock, and vacuum–single-photon mixed states; vacuum is also available as an explicit baseline.

The state controls and numerical controls serve different purposes. State controls change the physical state. Numerical controls change the representation, sampling, or classification resolution. Students should not use a numerical control to “repair” a physical conclusion; instead, they should establish convergence first and then interpret the converged result.

3. Photonic states represented in the notebook

3.1 Vacuum

The vacuum is the harmonic-oscillator ground state,

$$|0\rangle, \quad (1)$$

with zero mean field, one-half quantum of variance in each dimensionless quadrature, and a Gaussian Wigner function. It is both pure and Gaussian. In the Fock panel, all probability is at $n = 0$.

3.2 Coherent state

A coherent state is

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (2)$$

Its photon-number probabilities are Poisson distributed,

$$P_n = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}, \quad (3)$$

with mean photon number $\langle n \rangle = |\alpha|^2$. The Wigner function is a displaced Gaussian. Changing the phase of α rotates the displacement in phase space but does not change Gaussianity.

An important numerical lesson is that the exact coherent state has infinite Fock support. Any finite-dimensional calculation represents a truncation or finite-basis approximation. The user-controlled Hilbert-space dimension allows the student to verify when that approximation is adequate.

3.3 Squeezed vacuum

For squeezing parameter $\zeta = r e^{i\phi}$, squeezed vacuum can be written as an even-Fock-state superposition,

$$|0, \zeta\rangle = \frac{1}{\sqrt{\cosh r}} \sum_{k=0}^{\infty} \frac{\sqrt{(2k)!}}{2^k k!} \left[-e^{i\phi} \tanh r \right]^k |2k\rangle. \quad (4)$$

Only even photon numbers occur. Despite this highly non-Poissonian Fock structure, squeezed vacuum is a Gaussian state: Gaussianity refers to the phase-space characteristic or Wigner function, not to the shape of the photon-number histogram.

For a principal squeezed quadrature,

$$V_{\text{sq}} = \frac{1}{2} e^{-2r}, \quad V_{\text{asq}} = \frac{1}{2} e^{2r}. \quad (5)$$

The UI uses positive squeezing magnitude in decibels,

$$S_{\text{dB}} = 10 \log_{10} \left(\frac{V_{\text{vac}}}{V_{\text{sq}}} \right) = \frac{20r}{\ln 10}, \quad (6)$$

so that

$$r = \frac{\ln 10}{20} S_{\text{dB}}. \quad (7)$$

For example, 6 dB corresponds to $r \approx 0.6908$.

3.4 Fock states

A Fock state $|n\rangle$ has a definite photon number. For $n \geq 1$, it is non-Gaussian and its Wigner function contains negative regions. The state is parity symmetric, so all odd central quadrature moments vanish. Non-Gaussianity is exposed by the even moments.

For $|1\rangle$ in the convention used here,

$$W_1(x, p) = \frac{1}{\pi} [2(x^2 + p^2) - 1] e^{-(x^2 + p^2)}, \quad (8)$$

with

$$W_1(0, 0) = -\frac{1}{\pi}. \quad (9)$$

This is a large, unambiguous negative value compared with numerical truncation artifacts in nominally Gaussian states.

3.5 Vacuum–single-photon mixture

The mixed state

$$\rho(p) = (1 - p)|0\rangle\langle 0| + p|1\rangle\langle 1| \quad (10)$$

is the most useful counterexample in the notebook. Its Wigner function is

$$W_p(x, p_q) = \frac{e^{-R^2}}{\pi} [1 - 2p + 2pR^2], \quad R^2 = x^2 + p_q^2, \quad (11)$$

where p_q denotes the phase-space momentum coordinate to avoid confusion with the mixture weight p .

At the origin,

$$W_p(0, 0) = \frac{1 - 2p}{\pi}. \quad (12)$$

Therefore the Wigner function is nonnegative for $p \leq 1/2$ and becomes negative for $p > 1/2$. Yet the state is non-Gaussian for every $0 < p < 1$. This finite-parameter example demonstrates that a positive Wigner function does not imply Gaussianity for a general mixed state.

Its purity is

$$\text{Tr}(\rho^2) = (1 - p)^2 + p^2. \quad (13)$$

Purity is displayed because the relation between Wigner positivity and Gaussianity is stronger for pure states than for mixed states.

4. Quadratures and the Gaussian moment hierarchy

The notebook uses dimensionless quadratures

$$X = \frac{a + a^\dagger}{\sqrt{2}}, \quad P = \frac{a - a^\dagger}{i\sqrt{2}}, \quad (14)$$

so that

$$[X, P] = i, \quad \text{Var}_{|0\rangle}(X) = \text{Var}_{|0\rangle}(P) = \frac{1}{2}. \quad (15)$$

A rotated quadrature is

$$X_\theta = X \cos \theta + P \sin \theta. \quad (16)$$

The UI samples several directions over $0 \leq \theta < \pi$. For each direction, the central moments are

$$\mu_n(\theta) = \langle [X_\theta - \langle X_\theta \rangle]^n \rangle. \quad (17)$$

For a Gaussian probability distribution, all odd central moments vanish and all even central moments are determined by the variance:

$$\mu_{2k+1} = 0, \quad \mu_{2k} = (2k - 1)!! \mu_2^k. \quad (18)$$

The notebook therefore plots normalized residuals. For odd n ,

$$\delta_n = \frac{\mu_n}{\mu_2^{n/2}}, \quad (19)$$

and for even n ,

$$\delta_n = \frac{\mu_n}{(n - 1)!! \mu_2^{n/2}} - 1. \quad (20)$$

An exact Gaussian gives $\delta_n = 0$ for every tested order and direction. The reported Gaussianity score is

$$V_{\max} = \max_{\theta, n} |\delta_n(\theta)|. \quad (21)$$

Because the notebook tests only a finite set of moments and directions, this is a numerical Gaussian-consistency diagnostic rather than a formal proof of Gaussianity.

4.1 Why the non-Gaussian residuals become more negative

The residuals are not the moments themselves. For $|1\rangle$,

$$\delta_{2k} = \frac{2k+1}{3^k} - 1, \quad (22)$$

so $\delta_4 = -4/9$, $\delta_6 = -20/27$, $\delta_8 = -8/9$, and $\delta_{2k} \rightarrow -1$ as k increases. The interpretation is that the even moments become progressively smaller *relative to the Gaussian moments with the same variance*.

For the mixture $\rho(p)$,

$$\delta_{2k} = \frac{1+2kp}{(1+2p)^k} - 1, \quad (23)$$

which likewise approaches -1 for fixed $p > 0$. Odd residuals remain zero because both Fock states and this diagonal mixture are parity symmetric.

5. Wigner functions, positivity, and purity

The Wigner function is a quasiprobability distribution on phase space. In the convention used by the notebook, the vacuum Wigner function is

$$W_0(x, p) = \frac{1}{\pi} e^{-(x^2+p^2)}. \quad (24)$$

It integrates to unity but need not remain nonnegative for arbitrary quantum states. Negative regions are a strong nonclassical signature and, in several continuous-variable computational settings, Wigner negativity functions as a resource for escaping efficient classical simulation [6].

For pure continuous-variable states, Hudson's theorem states that a pure state has an everywhere non-negative Wigner function if and only if it is Gaussian [2]. The theorem does not extend in the same form to mixed states. The vacuum–single-photon mixture in Sec. 3.5 is deliberately included to make this distinction visible.

The simulator therefore reports three separate quantities:

Diagnostic	Question answered
Gaussian moment test	Is the sampled moment hierarchy consistent with a Gaussian state?
Wigner positivity	Is any negative Wigner value resolved at the selected cutoff?
Purity $\text{Tr}(\rho^2)$	Is the state pure or mixed?

No one of these three should be substituted for the other two.

6. Numerical controls and what they mean

6.1 Hilbert-space dimension N

The coherent and squeezed states have infinite Fock support. Numerically, they are represented in a finite space $|0\rangle, \dots, |N-1\rangle$. The UI therefore makes N a user parameter rather than silently choosing it. Increase N until the plotted observables and classification diagnostics stop changing appreciably.

High-order moments are especially sensitive to truncation. A state can have a tiny omitted probability tail while its eighth moment is still poorly converged because high moments weight large photon numbers strongly. Strong squeezing is therefore an effective convergence exercise: with insufficient N , a nominally Gaussian squeezed state can show spurious Wigner ripples and a large apparent eighth-order moment violation.

6.2 Wigner-negativity cutoff

A numerical Wigner array rarely gives exact zeros. The simulator defines resolved negativity through

$$W_{\min} < -W_{\text{cut}}. \quad (25)$$

The default is

$$W_{\text{cut}} = 10^{-7}. \quad (26)$$

The raw value of $W_{\min}$ is always reported. The cutoff is therefore not a modification of the Wigner function; it is a numerical decision threshold. A student should first establish basis and grid convergence, then judge whether a small negative value is physically resolved.

6.3 Phase-space extent and grid density

The phase-space extent controls the finite window over which $W(x, p)$ is evaluated, while the grid density controls spatial sampling. Too small an extent clips displaced or anti-squeezed tails. Too coarse a grid can miss narrow features or bias numerical integrals. Convergence should be checked by changing one parameter at a time.

6.4 Quadrature directions

The moment test samples X_θ over multiple directions. A single direction can miss anisotropic or correlation-dependent structure. Increasing the number of directions tests whether the Gaussianity conclusion is robust to phase-space rotation.

6.5 Moment-axis range

The default moment plot uses at least $[-1, +1]$. This avoids visually magnifying roundoff-level residuals such as 10^{-12} into apparently large deviations. The user can zoom the axis deliberately when studying numerical precision.

7. Suggested classroom workflow

A 60–90 minute guided activity can proceed as follows.

1. Start with vacuum at the default numerical settings and identify the three baseline signatures.
2. Compare coherent and squeezed states. Ask why their Fock histograms differ strongly even though both are Gaussian.
3. Select $|1\rangle$ and locate the negative Wigner region and the even-order moment violations.
4. Select the mixed state and vary p through $1/2$. Record what changes at the Wigner transition and what does not.
5. Increase squeezing to roughly 9–10 dB and deliberately lower N . Diagnose the resulting artifact, then restore convergence by increasing N .
6. Finally vary the Wigner cutoff and discuss why a numerical definition of “negative” requires a resolution statement.

The instructor should emphasize that the simulator is not a black-box classifier. The controls expose the assumptions behind the classification so that students can test whether a conclusion is stable.

8. Student exercises

Exercise 1 – Vacuum as the Gaussian reference

Select **Vacuum**. Use at least four quadrature directions.

- (a) Record the Fock probabilities, purity, $W_{\min}$, and maximum moment violation.
- (b) Verify that the Wigner function is rotationally symmetric.
- (c) Change the moment y-axis range from the default to a very small range. What do the newly visible residuals represent?

Exercise 2 – Coherent-state displacement

Select **Coherent**.

- (a) Set $|\alpha| = 1$ and rotate the coherent phase through 0° , 90° , 180° , and 270° . Describe the Wigner motion.
- (b) Increase $|\alpha|$ and compare the Fock histogram with the phase-space width.
- (c) Explain why the Wigner width does not increase with $|\alpha|$ even though the mean photon number does.

Exercise 3 – Squeezing in dB

Select **Squeezed vacuum**.

- (a) Enter 3, 6, and 9 dB and record the displayed r values.
- (b) Verify that only even Fock states are populated.
- (c) Rotate the squeezing phase and describe how the Wigner ellipse changes.
- (d) Explain why a non-Poissonian Fock distribution does not imply a non-Gaussian state.

Exercise 4 – Fock-state non-Gaussianity

Select **Fock state** and begin with $n = 1$.

- (a) Record $W_{\min}$ and compare it with $-1/\pi$.
- (b) Which moment orders show violations? Which remain zero?
- (c) Repeat for $n = 2$ and $n = 3$. How does the Wigner structure change?
- (d) Explain why parity symmetry can make all odd moments vanish without making the state Gaussian.

Exercise 5 – Positive-Wigner non-Gaussian state

Select **Vacuum / single-photon mixture**.

- (a) Set $p = 0.17$. Is the state Gaussian? Is its Wigner function positive? Is it pure?
- (b) Increase p in steps toward 0.50. Track $W(0, 0)$ and the moment residuals.
- (c) Move just above $p = 0.50$. Which classifier changes first?
- (d) Explain why the result does not contradict Hudson's theorem.

Exercise 6 – Locate the finite Wigner-negativity transition

Use the mixture and determine the transition value of p to within ± 0.01 using only the simulator. Then derive the transition analytically from $W_p(0, 0)$ and compare.

Exercise 7 – Hilbert-space convergence at strong squeezing

Select squeezed vacuum near 9–10 dB.

- (a) Begin with $N = 20$ or 30 and record $W_{\min}$ and the residuals through order eight.
- (b) Increase N systematically. At what value do the qualitative classification and the numerical diagnostics stabilize?
- (c) Which moment order is most sensitive to insufficient N ? Explain why.
- (d) State a convergence criterion in your own words that you would be willing to report in a research calculation.

Exercise 8 – What does “negative” mean numerically?

With a converged state, vary the Wigner cutoff from 10^{-4} to 10^{-12} .

- (a) Find a case where the YES/NO Wigner-positive label changes while the raw $W_{\min}$ does not.
- (b) Explain why the cutoff changes the interpretation but not the computed Wigner function.
- (c) Compare a tiny negative value from numerical truncation with the $|1\rangle$ value near -0.318 . Which is physically convincing and why?

Exercise 9 – Direct state comparison

Select **All 4 comparison states** and use common numerical settings.

- (a) Identify one visual signature that distinguishes each state family.
- (b) Identify one pair of states that cannot be reliably distinguished by Wigner positivity alone.
- (c) Identify one pair that cannot be reliably distinguished by the Fock histogram shape alone.
- (d) Write a three-sentence classification protocol using Wigner positivity, moment residuals, and purity together.

9. Notes for instructors

The strongest teaching moments occur when a simple visual rule fails. Three failures are particularly useful:

1. squeezed vacuum has a highly structured even-photon Fock distribution but remains Gaussian;
2. the mixed vacuum–single-photon state can be non-Gaussian while its Wigner function remains nonnegative;
3. insufficient Hilbert-space dimension can create apparent negativity or high-order moment violations in a state that should be Gaussian in the infinite-dimensional limit.

Students should be asked to distinguish *state physics* from *numerical evidence*. A statement such as “the Wigner function is negative” is incomplete in a numerical calculation unless the student can also state the observed minimum, the basis size, the phase-space grid, and the numerical cutoff used for classification.

10. Software and reproducibility

The notebook is implemented in Python/Jupyter and uses QuTiP for quantum-optical state and Wigner-function machinery [7, 8]. The validated development version contains unit tests for analytic Fock coefficients, squeezed-state parity and recurrences, quadrature variances, Gaussian moment identities, analytic Wigner functions, the $p = 1/2$ mixture threshold, Wigner negative volume, cutoff propagation, plotting logic, squeezing dB conversion, user-defined Hilbert dimension, and UI state selection.

For dissemination, the recommended workflow is:

1. provide the notebook and this teaching note in the same GitHub repository;
2. list Python, NumPy, Matplotlib, ipywidgets, and QuTiP as dependencies;
3. preserve the validated default parameters in the distributed notebook;
4. ask students to save or report parameter values used for any quantitative conclusion.

References

References

- [1] E. Wigner, “On the Quantum Correction For Thermodynamic Equilibrium,” *Phys. Rev.* **40**, 749 (1932).
- [2] R. L. Hudson, “When is the Wigner quasi-probability density non-negative?,” *Rep. Math. Phys.* **6**, 249–252 (1974), doi:10.1016/0034-4877(74)90007-X.
- [3] U. Leonhardt, *Measuring the Quantum State of Light* (Cambridge University Press, Cambridge, 1997).
- [4] C. C. Gerry and P. L. Knight, *Introductory Quantum Optics* (Cambridge University Press, Cambridge, 2005).
- [5] C. Weedbrook, S. Pirandola, R. García-Patrón, N. J. Cerf, T. C. Ralph, J. H. Shapiro, and S. Lloyd, “Gaussian quantum information,” *Rev. Mod. Phys.* **84**, 621 (2012).
- [6] A. Mari and J. Eisert, “Positive Wigner Functions Render Classical Simulation of Quantum Computation Efficient,” *Phys. Rev. Lett.* **109**, 230503 (2012), doi:10.1103/PhysRevLett.109.230503.
- [7] J. R. Johansson, P. D. Nation, and F. Nori, “QuTiP: An open-source Python framework for the dynamics of open quantum systems,” *Comput. Phys. Commun.* **183**, 1760–1772 (2012), doi:10.1016/j.cpc.2012.02.021.

- [8] J. R. Johansson, P. D. Nation, and F. Nori, “QuTiP 2: A Python framework for the dynamics of open quantum systems,” *Comput. Phys. Commun.* **184**, 1234–1240 (2013), doi:10.1016/j.cpc.2012.11.019.

Appendix A. Expected outcomes and explanations

This appendix is intended for instructors, self-study, or post-activity checking. Numerical values can vary slightly with Hilbert-space dimension, phase-space window, grid density, quadrature sampling, and Wigner cutoff.

A.1 Exercise 1 – Vacuum

Expected outcome: $P_0 = 1$, purity = 1, Gaussian classification YES, and Wigner-positive classification YES. The Wigner function is circularly symmetric and centered at the origin. All Gaussian residuals should be zero to numerical precision. If the moment axis is zoomed to 10^{-12} – 10^{-14} scales, small nonzero values may appear; these are numerical roundoff, not evidence that the vacuum is physically non-Gaussian.

A.2 Exercise 2 – Coherent state

Expected outcome: changing the coherent phase rotates the center of the Wigner Gaussian while preserving its width and shape. Increasing $|\alpha|$ moves the center farther from the origin and broadens the Poisson photon-number distribution because $\langle n \rangle = |\alpha|^2$. The quadrature variance remains at the vacuum value in every direction. Thus a larger mean energy does not require a broader Wigner Gaussian; the energy increase is primarily displacement energy.

A.3 Exercise 3 – Squeezed vacuum

Expected r values are approximately 0.3454, 0.6908, and 1.0362 for 3, 6, and 9 dB. Only even Fock probabilities are nonzero. The Wigner function remains Gaussian but becomes elliptical; changing the squeezing phase rotates the ellipse. The key lesson is that Gaussianity is a phase-space property, not Gaussianity of the photon-number histogram.

A.4 Exercise 4 – Fock states

For $|1\rangle$, $W_{\min}$ should be close to $-1/\pi \approx -0.31831$ when the origin lies on the phase-space grid. Odd central moments vanish by parity, while even Gaussian residuals are nonzero. In the implemented normalization,

$$\delta_4 = -\frac{4}{9} \approx -0.4444, \quad \delta_6 = -\frac{20}{27} \approx -0.7407, \quad \delta_8 = -\frac{8}{9} \approx -0.8889. \quad (27)$$

Higher Fock states produce additional radial oscillations in $W(x, p)$ and remain non-Gaussian.

A.5 Exercise 5 – Positive-Wigner non-Gaussian mixture

At $p = 0.17$, the state is mixed, non-Gaussian, and Wigner-positive. Its purity is

$$(0.83)^2 + (0.17)^2 = 0.7178. \quad (28)$$

The Wigner value at the origin is

$$W(0, 0) = \frac{0.66}{\pi} \approx 0.2101 > 0. \quad (29)$$

The non-Gaussianity appears in the even moment hierarchy. This does not contradict Hudson's theorem because the state is not pure.

A.6 Exercise 6 – Mixture transition

The exact transition occurs at $p = 1/2$ because

$$W(0, 0) = \frac{1 - 2p}{\pi}. \quad (30)$$

For $p < 1/2$, the Wigner function is nonnegative. At $p = 1/2$, it touches zero at the origin. For $p > 1/2$, a negative region appears. Gaussianity does *not* suddenly change at $p = 1/2$; the state was already non-Gaussian for any $p > 0$.

A.7 Exercise 7 – Hilbert-space convergence

At strong squeezing, low N can create two conspicuous artifacts: a small negative Wigner ripple and a large high-order moment residual, often first visible at order eight. As N increases, both should shrink and stabilize. The exact convergence value depends on squeezing and the selected observables, so there is intentionally no universal “correct” N . A defensible convergence statement is, for example: “Increasing N further changed $W_{\min}$ and all reported moment residuals by less than the chosen reporting tolerance.”

A.8 Exercise 8 – Wigner cutoff

Changing the cutoff can change only the YES/NO interpretation, not the raw $W_{\min}$. If a state has $W_{\min} = -6 \times 10^{-8}$, it is classified as unresolved negativity with a 10^{-7} cutoff but as resolved negativity with a 10^{-8} cutoff. This is a numerical-resolution statement. By contrast, $W_{\min} \approx -0.318$ for $|1\rangle$ is so far from zero that reasonable numerical cutoffs do not affect the physical conclusion.

A.9 Exercise 9 – Direct comparison

A useful classification protocol is: first establish numerical convergence; second inspect the Wigner function and record $W_{\min}$ relative to the stated cutoff; third inspect the Gaussian moment residuals over multiple quadrature directions; finally use purity to determine whether pure-state results such as Hudson’s theorem can be applied. The coherent and squeezed states are both Gaussian and Wigner-positive despite very different Fock distributions. The positive-Wigner mixture shows why Wigner positivity alone is insufficient for mixed-state Gaussianity classification.

Appendix B. Instructor discussion prompts

1. Why does infinite Fock support not automatically imply non-Gaussianity?
2. Why can an exact Gaussian state acquire apparent Wigner negativity in a finite numerical basis?
3. Why are high-order moments more sensitive to truncation than normalization or mean photon number?
4. What additional evidence would you require before claiming experimentally observed Wigner negativity?
5. In what sense is Wigner negativity a resource, and why should that statement not be generalized beyond the assumptions of a particular computational model?